

國立臺灣大學 期中考試答案卷

National Taiwan University Midterm/Final Examination Answer Sheet

課程編號

Course no.

科目

Course title

考試日期

Date

從此處開始寫起。試卷用紙務須節用。非經主試認可不得續用其他紙張作答。

Please write from here.

學院 資管

College

學系

Department

學號

Student ID no.

姓名

Name

組

Division

年級

Year

記分	教師簽名或蓋章
Score	Lecturer's signature

91

- For busy-waiting, CPU has to go in a loop until it encounters busy() to do some operations, CPU will be idle for a large amount of time. For interrupt-based operations, CPU keeps executing until an interrupt occurred. CPU stop the current job and start executing another job, which will always make CPU busy.
- The device controller stores the data in its register. When the data is ready, the hardware sends a signal via the interrupt-request line. CPU catches the signal and save the current state, transfer into a generic routine. CPU uses the interrupt vector to find the job that the interrupt wants to do, and calls the interrupt-specific handler to deal with it. After executing the interrupt, CPU resumes to what it was doing before the hardware interrupt.

The device drivers send request to the device controllers. If it gets the request, the device controller asks the devices to do some I/O operations.

The dual-mode operation means that operations can be execute in user mode and kernel mode. Some operations that do harm will only be execute in the kernel mode. User mode will be switched into kernel mode when an interrupt, or a trap occurs. The dual-mode operation ensures that users won't access the kernel of the operation system so it could prevent errant users.

4.

+10 (a) In the user mode, an API function invokes a software interrupt and a system call number, which is stored in EAX register, to be sent to CPU. Use the system call number and search the table of code pointer to find the addresses for the system call. CPU switches to kernel mode and execute the operations. After the operations for the system call are done, CPU switches back to user mode.

+5 (b) API functions have better portability. They can be compiled and run on different systems, if that system support these API functions. Also, API functions are much more easier for the users to handle than making system call directly.

(c) One of the methods is to use parameters marshalling, which is to transfer data to an appropriate type that could go through transmission.

5. The modular kernel approach has higher efficiency since it doesn't have to go through so many layers and can still perform most of the proper execution of the operating system.

(a) ready

(b) ready

(c) running

(d) running

(e) waiting

(f) ready

(g). ready